

CoSE-RAG: Connectivity- and Sufficiency-Aware Evidence Retrieval for Complex Document Question Answering

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Abstract—Complex document question answering usually requires retrieving from long documents a set of evidence that can jointly support the answer. Relevant evidence may be distributed across different paragraphs, sections, tables, and figure captions and connected through local context, shared entities, or section hierarchies; retrieval-augmented generation methods based on fixed-length segmentation and a fixed top-k can easily miss key evidence, break evidence connections, or introduce redundant noise. To this end, we propose Connectivity- and Sufficiency-aware Evidence Retrieval-Augmented Generation (CoSE-RAG). The method constructs a multi-view document evidence graph containing atomic evidence blocks, structural patches, semantic entities, and hierarchical summaries, and uses contextual, semantic, and hierarchical edges to represent cross-block, cross-entity, and local-global relations. It performs multi-granularity seed retrieval according to the question requirements and expands and connects relevant evidence through typed Personalized PageRank (PPR) and constrained shortest paths. Finally, it constructs the generation context under dual budgets on the number of blocks and tokens, uses sufficiency diagnosis to trigger targeted re-retrieval, and outputs traceable supporting evidence from the generation context. On the 281 papers and 1005 questions in Qasper, CoSE-RAG achieves an Answer F1, Evidence Precision, Evidence Recall, and Evidence F1 of 45.44%, 47.34%, 57.26%, and 48.79%, respectively, with all four metrics being the best among the compared methods. On a fixed sample of 1000 HotpotQA distractor validation questions, its answer Exact Match (EM), Answer F1, Supporting Fact Recall, and Supporting Fact F1 reach 62.60%, 76.36%, 80.06%, and 67.52%, respectively, with all four metrics being the best among the compared methods. The results show that CoSE-RAG can simultaneously improve answer quality, evidence coverage, and evidence alignment quality in complex document question answering.

Index Terms—retrieval-augmented generation, complex document question answering, evidence retrieval, graph retrieval, large language models

I. INTRODUCTION

In recent years, large language models (LLMs) have made significant progress in tasks such as question answering and

knowledge reasoning. However, in the absence of reliable external context, they may still be limited by knowledge time-liness, domain coverage, and factual consistency. Retrieval-augmented generation (RAG) retrieves relevant content from external knowledge sources at inference time and uses it as the generation context, providing an updatable and verifiable information basis for knowledge-intensive question answering [1–3]. Nevertheless, retrieving relevant passages is not equivalent to obtaining evidence sufficient to answer the question.

Complex documents, such as scientific papers, technical reports, and industry specifications, contain section hierarchies, tables, figure captions, term definitions, and cross-paragraph references, and the information required by a question is often distributed across multiple locations. For example, question answering over scientific papers may simultaneously depend on method descriptions, experimental settings, and result explanations, whereas multi-hop question answering requires connecting different entities or events. Related studies show that document structure and cross-region relations are important information for complex document understanding [4]. Meanwhile, even when a model has a long context window, key information located in the middle of the context or surrounded by distracting content may still be difficult to use reliably [5]. Therefore, the key to complex document question answering is not to expand the input unconditionally, but to construct, within a limited budget, an evidence set that covers the question requirements, preserves internal connections, and controls noise.

Traditional RAG generally performs sparse or dense retrieval over fixed segments and then returns a fixed number of high-scoring passages. This process has three limitations for complex documents: fixed boundaries may disrupt the structural relations among sections, figure or table descriptions, and adjacent paragraphs; ranking individual passages by relevance makes it difficult to preserve the intermediate

connections required by comparison, multi-hop, and global questions; and a uniform top- k cannot determine whether the current evidence is sufficient according to question difficulty. Hierarchical retrieval, knowledge-graph enhancement, and adaptive retrieval alleviate some of these problems, respectively. KG²RAG performs knowledge-graph expansion and text organization from semantic seeds, HippoRAG 2 combines passage nodes with Personalized PageRank (PPR), GeAR supports multi-step retrieval through graph expansion, and S2G-RAG explicitly judges evidence sufficiency and outputs structured information gaps [6–9]. These methods indicate that both graph structures and sufficiency control are beneficial for multi-evidence question answering, but they mainly target knowledge graphs, multi-hop corpora, or general iterative retrieval. Context, shared entities, and section hierarchies within complex documents jointly form heterogeneous evidence relations, and the methods above have not yet jointly modeled these relations under unified provenance mapping and explicit budget constraints.

We propose Connectivity- and Sufficiency-aware Evidence Retrieval-Augmented Generation (CoSE-RAG), which transforms the retrieval objective from ranking independent passages into constructing an evidence set aware of connectivity and sufficiency. The method first builds a multi-view document evidence graph from atomic evidence, local patches, semantic entities, and hierarchical nodes, while retaining provenance mappings from auxiliary nodes to locations in the source document. It then adjusts the propagation weights of contextual, semantic, and hierarchical relations according to the question requirements and expands candidate evidence through multi-granularity seeds, typed graph propagation, and constrained path connection. Finally, it selects the generation context under dual budgets on the number of blocks and tokens, and uses sufficiency diagnosis to locate missing requirements and trigger a limited number of rounds of targeted supplementary retrieval. To ensure the controllability and reproducibility of closed-loop retrieval, re-retrieval under insufficient evidence performs targeted expansion only along predefined node and relation types.

The contributions of this paper are threefold. First, we propose a multi-view document evidence graph that preserves provenance mappings and organizes atomic content, local structures, semantic anchors, and section hierarchies in a unified representation, while distinguishing atomic nodes that can directly support an answer from auxiliary nodes used only for retrieval expansion. Second, we propose a typed evidence retrieval method driven by question requirements, which constructs a compact and internally connected evidence set through adaptive relation propagation, constrained connection, and budgeted selection. Third, we design controlled evidence sufficiency diagnosis and targeted re-retrieval mechanisms that supplement candidates along specified relations when evidence is missing and constrain the final supporting evidence to originate from the context actually used by the generator and to be traceable to the source document.

II. RELATED WORK

A. Retrieval-Augmented Generation and Adaptive Retrieval

Retrieval-augmented generation introduces external knowledge through indexing, retrieval, and generation, while retrieval quality and context organization jointly affect the answer [1–3, 10]. BM25, dense retrieval, Reciprocal Rank Fusion (RRF), and BERT reranking represent standard sparse, dense, fusion, and reranking paradigms [11–14]. ColBERTv2, Atlas, and REPLUG improve semantic matching through multi-vector interaction or retrieval-augmented modeling [15–17]; IRCot and ReAct interleave retrieval with reasoning or actions, whereas FLARE, Self-RAG, and S2G-RAG control retrieval and iteration according to the reasoning state, generation state, or structured information gaps [9, 18–21]. These methods improve recall and iteration timing, but generally do not explicitly preserve section membership, adjacent content, and figure and table relations in complex documents.

B. Complex Document Understanding and Long-Context Question Answering

Complex documents contain titles, paragraphs, tables, figure captions, and layout elements, and simple flattening loses section membership and cross-region correspondences [4]. LayoutLMv3, Donut, UDOP, and DocLLM enhance document understanding by jointly modeling vision, text, and layout [22–25]. However, long-context models may still overlook information in the middle of the context or surrounded by noise [5], and LongBench v2, MMLongBench-Doc, and DocBench also show that long documents, multi-page structures, and unanswerable questions remain challenging [26–28]. This paper focuses on evidence retrieval after document parsing and organizes traceable textual evidence under a given budget, rather than processing visual semantics in raw images.

C. Hierarchical and Graph-Enhanced Retrieval and Evidence Sufficiency

RAPTOR constructs a tree index through recursive clustering and summarization, while LongRAG retrieves longer units to reduce retrieval fragmentation [29, 30]. Research integrating knowledge graphs (KGs) with LLMs uses entity relations to support structured reasoning [31–33]; G-Retriever, the HippoRAG series, KG²RAG, and GeAR further introduce graph question answering, PPR-based associative retrieval, or graph expansion into RAG [6–8, 34, 35]. These methods demonstrate the value of graph structures for cross-passage retrieval, but generally center on knowledge entities or general corpus relations and lack unified provenance mapping and typed propagation for atomic content, local patches, and section hierarchies in complex documents.

ALCE, RARR, and generative-search evaluation examine the correspondence between generated claims and external evidence [36–38], while RAGAS, ARES, and RAGChecker evaluate context relevance, answer faithfulness, and fine-grained errors [39–41]. Sufficient Context and SURE-RAG model evidence sufficiency from the question–context and

answer-level set perspectives, respectively [42, 43], and S2G-RAG and GeAR use it to control subsequent retrieval or graph expansion [8, 9]. Unlike these studies, this paper uses pre-generation diagnosis to control typed re-retrieval over a multi-view document evidence graph and jointly models the connectivity, coverage, and source traceability of complex-document evidence through dual-budget selection.

In summary, existing studies improve semantic matching, long-document representation, graph expansion, adaptive retrieval, or evidence evaluation, but usually address these aspects separately. Existing methods generally do not jointly preserve provenance across atomic content and document structure, adapt typed relation propagation to question requirements, and construct a sufficient evidence set under explicit block and token budgets. These gaps motivate the proposed methodology.

III. PROBLEM DEFINITION AND METHOD OVERVIEW

Given a complex document collection $D = \{d_i\}_{i=1}^n$ and a user question q , we study how to construct, within a limited context budget, an evidence set that can jointly support the answer. Unlike independent-passages ranking, this task requires not only that the evidence be relevant to the question, but also that multiple pieces of evidence cover the question requirements, preserve necessary connections, and be traceable to the source documents.

The system indexes D offline as a multi-view document evidence graph G_{CoSE} , constructs the final generation context C_{gen} online, determines whether it is sufficient, and generates an answer a and a supporting evidence set C_s , as formalized in Eq. (1):

$$\begin{aligned} G_{\text{CoSE}} &= \mathcal{I}(D), \\ (C_{\text{gen}}, y) &= \mathcal{R}_{\text{CoSE}}(q, G_{\text{CoSE}}; B_{\text{blk}}, B_{\text{tok}}, R_{\text{max}}), \\ (a, C_s) &= \mathcal{G}_{\theta}(q, C_{\text{gen}}, y), \quad C_s \subseteq C_{\text{gen}} \cap V_b^{\text{sup}}. \end{aligned} \quad (1)$$

Here, B_{blk} and B_{tok} are the upper limits on the number of evidence blocks and input tokens, respectively, and R_{max} is the maximum number of retrieval rounds; $y \in \{0, 1\}$ indicates evidence sufficiency at termination; and V_b^{sup} denotes the atomic evidence nodes allowed to support an answer directly. C_{gen} is the context actually received by the generator, whereas C_s is the subset cited by the final answer and validated for provenance. This inclusion relation ensures that the system does not retrospectively list content unused during generation as answer evidence.

The overall process is shown in Fig. 1. CoSE-RAG first parses controlled evidence requirements from the question and performs multi-granularity seed retrieval. It then adjusts the propagation weights of contextual, semantic, and hierarchical relations according to the requirements and expands candidates through graph propagation and constrained path connection. The budgeted selector constructs C_{gen} accordingly, and the sufficiency verifier determines its coverage and connectivity. If the evidence is insufficient, the diagnostic result triggers targeted re-retrieval only within predefined node and relation

types, after which propagation and budgeted selection are performed again. Once the evidence is sufficient or the iteration limit is reached, the generator outputs the answer and traceable supporting evidence from the final C_{gen} .

IV. THE PROPOSED METHODOLOGY

CoSE-RAG constructs an evidence set that can support the answer and remain traceable under the constraints of B_{blk} , B_{tok} , and R_{max} . The system builds the multi-view evidence graph offline and sequentially performs requirement parsing, seed retrieval, typed bridging, budgeted selection, and sufficiency verification online. When evidence is insufficient, it expands seeds only within predefined requirement fields, node types, and bridge types, without rewriting the question or introducing facts external to the documents.

A. Multi-View Document Evidence Graph

a) Nodes and Provenance Mapping: To represent local facts, cross-block relations, and section scopes in a unified manner, CoSE-RAG represents the documents as shown in Eq. (2):

$$\begin{aligned} G_{\text{CoSE}} &= (V, E, M), \\ V &= V_b \cup V_p \cup V_e \cup V_h, \\ E &= E_c \cup E_s \cup E_h. \end{aligned} \quad (2)$$

where V_b contains atomic nodes such as paragraphs, tables, figure captions, images, and formulas; V_p contains local patches composed of adjacent atomic blocks in the same section; V_e contains entity or term nodes; and V_h contains hierarchical nodes such as titles, section overviews, and document outlines. E_c , E_s , and E_h represent contextual, semantic, and hierarchical relations, respectively. An edge $e = (u, v, t_e, \rho, w_e)$ records the bridge type t_e , the fine-grained relation ρ , and the reliability prior w_e given by the graph-construction rule. Direct citations, figure-caption correspondences, and section membership have higher priors than weak term co-occurrence relations. $M : V \rightarrow 2^S$ stores mappings from nodes to sets of source locations, whose identifiers contain at least the document, section path, and original block. The mapping of an auxiliary node is obtained from the union of its source blocks and is projected back to atomic content through M . The final evidence is drawn only from $V_b^{\text{sup}} \subseteq V_b$, whose nodes have verifiable text and nonempty mappings.

b) Heterogeneous Document Content and Bridging Relations: Different types of content are uniformly converted into traceable text: tables retain their titles, headers, and rows; formulas retain nearby explanations; and figure captions are connected to their corresponding images. Images without parsable text serve only as structural anchors. Overlong content is segmented at sentence or table-row boundaries, local patches are combined in their original order, and section overviews use extractive content. Contextual and hierarchical edges are determined by document structure and section membership, respectively. Let $T(v)$ denote the term set after stop-word

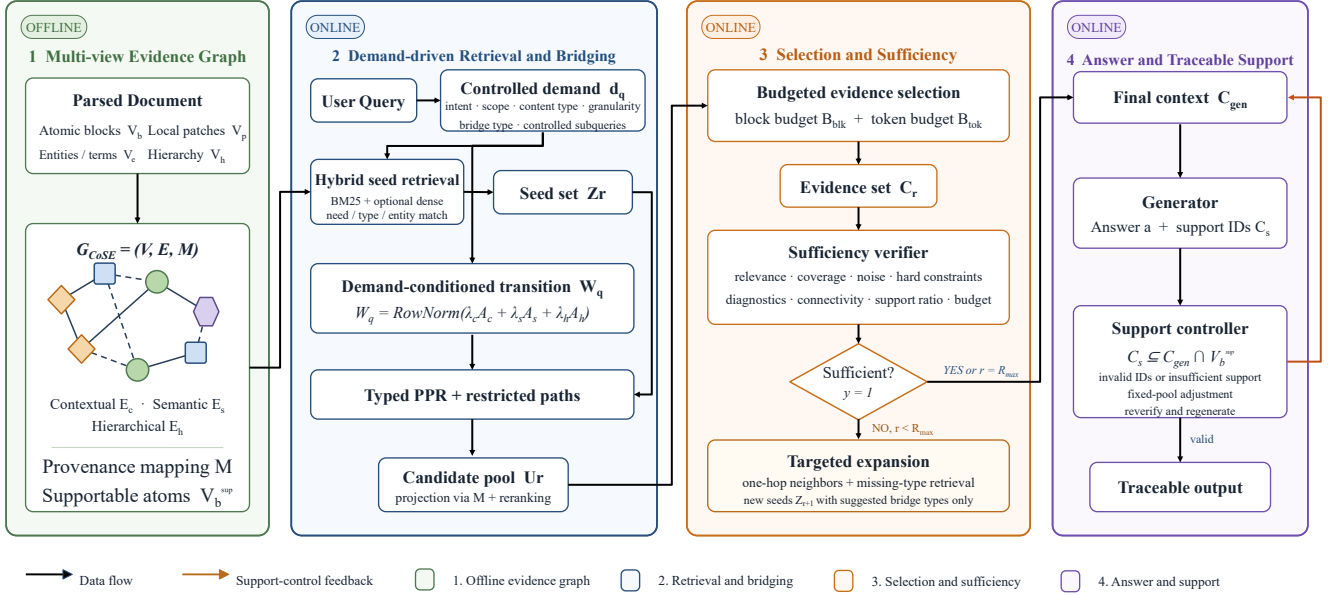


Fig. 1. Overall process of CoSE-RAG. Auxiliary nodes are used for retrieval and bridging, while the final supporting evidence is provided by atomic nodes traceable to the source documents.

filtering. Shared-term edges measure term overlap as shown in Eq. (3):

$$\text{ov}(u, v) = \frac{|T(u) \cap T(v)|}{|T(u) \cup T(v)|} \quad (3)$$

An edge is created only when the number of shared terms or the overlap ratio reaches its threshold; the value is set to 0 if the union is empty. This edge is used only for candidate expansion and does not represent logical entailment. Segmentation lengths, filtering rules, and edge weights are provided in the experimental settings.

B. Requirement-Aware Seed Retrieval

a) Controlled Evidence Requirements: The system first parses the question into the controlled evidence-requirement tuple defined in Eq. (4):

$$d_q = \Psi(q) = (\iota_q, s_q, \mathcal{O}_q, g_q, \mathcal{B}_q, \mathcal{Q}_q), \quad (4)$$

where ι_q , s_q , $\mathcal{O}_q$, g_q , $\mathcal{B}_q$, and $\mathcal{Q}_q$ denote the intent, scope, content types, granularity, preferred bridge types, and at most two controlled subqueries, respectively. Predefined mappings cause factual, comparison or multi-hop, figure/table, and global questions to favor contextual, semantic, figure/table-associated, and hierarchical evidence, respectively, and jointly control retrieval, propagation, paths, re-retrieval, and sufficiency constraints. The parser prioritizes high-confidence rules. When confidence is low, the LLM outputs a structured result within the same limited fields and falls back to the rules upon failure. Neither receives gold answers or gold evidence, nor do they generate free-form retrieval steps.

b) Multi-Granularity Hybrid Seeds: The system performs BM25 and optional dense retrieval over atomic blocks and auxiliary nodes, with the normalized channel scores fused as shown in Eq. (5):

$$\begin{aligned} h_q(v) &= \beta \widehat{s}_{\text{BM25}}(v) + (1 - \beta) \widehat{s}_{\text{dense}}(v), \\ s_0(v) &= \max\{h_q(v), b(v; d_q)\} + \chi(v, q), \end{aligned} \quad (5)$$

where $\beta \in [0, 1]$ is the sparse-dense fusion coefficient, $b(v; d_q)$ measures type matching, and $\chi(v, q)$ rewards entities or key terms in the question. When the dense channel is disabled, $h_q(v)$ reduces to BM25. $b(v; d_q)$ applies only to retrieved candidates, preventing nodes from being introduced unconditionally by type.

After merging and deduplicating candidates from all channels, the system retains the K_{seed} highest-scoring seeds according to Eq. (6):

$$Z_0 = \text{Top}_{K_{\text{seed}}}(\mathcal{R}(q, d_q, G_{\text{CoSE}}), s_0). \quad (6)$$

Atomic blocks provide answer content, local and hierarchical nodes delimit the scope, and entity nodes provide cross-location entry points. Z_0 serves only as the propagation starting point, while the final evidence is still determined by the budgeted selector.

C. Requirement-Driven Typed Evidence Bridging

a) Question-Adaptive Propagation: Let A_c , A_s , and A_h be the weighted adjacency matrices of the three edge types. The requirement d_q is mapped to nonnegative type weights $\lambda_c, \lambda_s, \lambda_h$ that sum to 1, yielding the transition matrix in Eq. (7):

$$W_q = \text{RowNorm}(\lambda_c A_c + \lambda_s A_s + \lambda_h A_h). \quad (7)$$

Fusion before normalization preserves the prior strengths of different relations; a node without outgoing edges retains a self-loop. Local or figure/table questions, multi-hop or comparison questions, and global questions increase the contextual, semantic, and hierarchical weights, respectively, and the mapping rules are fixed before evaluation.

The scores of the round- r seeds Z_r are normalized as $p_r^{(0)}$, and typed PPR is performed over W_q according to Eq. (8):

$$p_r^{(t+1)} = \alpha p_r^{(0)} + (1 - \alpha) W_q^\top p_r^{(t)}. \quad (8)$$

where $\alpha \in (0, 1)$ is the restart probability, and $p_r = p_r^{(T_p)}$ after T_p iterations. The question-adaptive W_q and the restart term jointly limit irrelevant diffusion. To prevent direct seeds from being diluted, we apply the correction in Eq. (9):

$$\bar{p}_r(v) = p_r(v) + \delta p_r^{(0)}(v) \mathbf{1}[v \in Z_r], \quad (9)$$

where $\delta \geq 0$ is the seed-preservation coefficient and $\mathbf{1}[\cdot]$ is the indicator function, which equals 1 when the condition holds and 0 otherwise. High-scoring propagated candidates and high-confidence atomic seeds are retained to obtain $U_{\text{ppr}}^{(r)}$.

b) Constrained Path Connection and Candidate Reranking: To complete connecting nodes, the system searches, over the bridge types specified by $\mathcal{B}_q$, for shortest paths between Z_r and high-scoring propagated candidates that contain intermediate nodes and have at most L_{path} hops. The connector set and resulting candidate pool are defined in Eq. (10):

$$\begin{aligned} \Pi_q^{(r)} &= \left\{ \pi(z, u) \mid z \in Z_r, u \in U_{\text{ppr}}^{(r)} \setminus Z_r, \right. \\ &\quad \left. 2 < |V(\pi)| \leq L_{\text{path}} + 1 \right\}, \\ U_r &= U_{\text{ppr}}^{(r)} \cup \bigcup_{\pi \in \text{First}_{K_{\text{path}}}(\Pi_q^{(r)})} V(\pi). \end{aligned} \quad (10)$$

The search is performed on an undirected projected graph filtered by bridge type, with stable sorting by hop count, edge reliability, and node identifier in sequence. $\text{First}_{K_{\text{path}}}$ takes the first K_{path} nonduplicate paths. Intermediate nodes receive decayed scores according to their path positions. A path represents only a retrieval connection and is not regarded as a verified reasoning chain.

Auxiliary nodes are projected with decay through M to their source atomic nodes, while a small number of auxiliary nodes are retained. Let $s_\pi(v)$ and $s_M(v)$ denote the path-position score and provenance-projection score, respectively. After normalizing all terms, the aggregate bridge score is computed by Eq. (11):

$$s_{\text{bridge}}(v) = \gamma_p \hat{p}_r(v) + \gamma_\pi \hat{s}_\pi(v) + \gamma_M \hat{s}_M(v), \quad (11)$$

where $\gamma_p, \gamma_\pi, \gamma_M \geq 0$ and their sum is 1. A fixed number of high-scoring atomic blocks $U_{\text{rank}}^{(r)}$ further incorporate reranking scores according to Eq. (12):

$$r_q(v) = \begin{cases} (1 - \eta) \hat{s}_{\text{bridge}}(v) + \eta \hat{s}_{\text{rerank}}(v), & v \in U_{\text{rank}}^{(r)}, \\ \hat{s}_{\text{bridge}}(v), & \text{otherwise.} \end{cases} \quad (12)$$

where the two scores are normalized separately and $\eta \in [0, 1]$ is a fusion coefficient fixed before evaluation. Candidates outside the set retain their graph-bridging scores.

D. Sufficiency-Driven Selection and Closed-Loop Control

a) Greedy Selection under Dual Budgets: Under budgets on the number of blocks and tokens, the system iteratively selects the candidate with the largest marginal gain, as formalized in Eq. (13):

$$\begin{aligned} v^* &= \arg \max_{v \in \mathcal{F}(U_r, C_k)} \Delta(v \mid C_k), \\ \Delta(v \mid C_k) &= r_q(v) + \omega_n \Phi_{\text{need}}(v \mid C_k, d_q) \\ &\quad + \omega_l \Phi_{\text{link}}(v \mid C_k) + \omega_d \Phi_{\text{div}}(v \mid C_k) \\ &\quad + \omega_o \Phi_{\text{role}}(v) - \omega_r \text{Red}(v, C_k) \\ &\quad - \omega_b \text{Cost}(v). \end{aligned} \quad (13)$$

Here, $\mathcal{F}(U_r, C_k)$ is the set of unselected candidates whose addition still satisfies both budgets. Φ_{need} , Φ_{link} , Φ_{div} , and Φ_{role} measure requirement supplementation, direct connection, newly added content or relations, and node role, respectively, while Red and Cost penalize textual redundancy and normalized token cost. The nonnegative weights $\omega_n, \omega_l, \omega_d, \omega_o, \omega_r, \omega_b$ are fixed before evaluation, and an additional upper limit is placed on the number of auxiliary nodes. Selection stops when either budget is reached, candidates are exhausted, or no feasible candidate remains. The round- r result is denoted by C_r , and below C is used generically for the C_r to be verified.

b) Sufficiency Diagnosis and Targeted Re-Retrieval: Based on the question requirements, current evidence, and its internal relations, the verifier produces the tuple in Eq. (14):

$$\Omega(q, d_q, C, E[C]) = (y, \mathcal{M}, \mathcal{L}, \xi_{\text{next}}, \mathbf{z}), \quad (14)$$

where y is the acceptance indicator; $\mathcal{M}$, $\mathcal{L}$, and ξ_{next} are the missing requirements, suggested bridge types, and controlled action, respectively; and $E[C]$ is the induced edge set. $\mathbf{z}$ summarizes relevance, coverage, connectivity, the proportion of atomic evidence, noise, and budget usage. The budgets are already guaranteed by the selector and therefore no acceptance threshold is set for them.

The basic acceptance condition of the rule-based verifier is given in Eq. (15):

$$\begin{aligned} y_{\text{rule}} &= \mathbf{1} \left[\text{Rel}(q, C) \geq \tau_r \wedge \text{Cov}(d_q, C) \geq \tau_c \right. \\ &\quad \left. \wedge \text{Noise}(C) \leq \tau_n \wedge H(d_q, C) \right]. \end{aligned} \quad (15)$$

where $\tau_r, \tau_c, \tau_n \in [0, 1]$ are the relevance, coverage, and noise thresholds, respectively, and $H(d_q, C)$ checks hard constraints such as content type, multi-hop connectivity, and section coverage for global questions. We set $y = y_{\text{rule}}$. The optional structured verifier only refines $\mathcal{M}$, $\mathcal{L}$, and ξ_{next} ; it neither changes the acceptance condition nor receives gold answers or gold evidence.

If $y = 0$ and $r + 1 < R_{\text{max}}$ (the initial retrieval is counted as round 1, i.e., $r = 0$), the system supplements the seeds from the controlled one-hop neighborhood of C_r and retrieval

results for the missing types. For $v \in V \setminus (Z_r \cup C_r)$, the targeted expansion score is defined in Eq. (16):

$$\begin{aligned}\tilde{s}_{r+1}(v) &= \max\{e_r(v), m_r(v)\}, \\ e_r(v) &= \kappa_e \max_{\substack{u \in C_r \\ (u,v) \in E_{\mathcal{L}}}} w_{uv}, \\ m_r(v) &= \kappa_m s_{\text{type}}(v, q) \\ &\quad \cdot \mathbf{1}[v \in \mathcal{T}(\mathcal{M}, \xi_{\text{next}})].\end{aligned}\tag{16}$$

where e_r and m_r are the neighborhood-expansion and missing-type retrieval scores, respectively, and the nonnegative coefficients κ_e, κ_m control their scales. $E_{\mathcal{L}}$ contains only the suggested bridge types, w_{uv} is the edge prior, $s_{\text{type}}(v, q)$ is the targeted type-retrieval score, and $\mathcal{T}(\mathcal{M}, \xi_{\text{next}})$ returns node types matching the missing requirements and action. When there is no connected neighbor, $e_r(v) = 0$. Source atomic blocks are first supplemented for auxiliary candidates through M ; at most K_{seed} new nodes are then retained and merged with Z_r into Z_{r+1} , followed by propagation, connection, reranking, and selection again. The process terminates early if no new atomic candidate is found in two consecutive rounds or if neither the evidence set nor its score improves.

c) Answer Generation and Supporting-Evidence Control: After retrieval terminates, the selection set from the final round is taken as C_{gen} . If it contains no supporting atomic node, the system returns insufficient evidence. Based only on this context, the generator outputs the answer and supporting-block identifiers, and verifies whether the identifiers belong to $C_{\text{gen}} \cap V_b^{\text{sup}}$. When citations are empty or invalid, valid citations are insufficient for multi-hop or comparison questions, or $y = 0$, the controller supplements missing evidence or removes redundant nodes within the fixed candidate pool and the original budgets. Once the context changes, verification and generation must be performed again. The final C_s retains only valid regenerated identifiers and therefore satisfies (1). Unanswerable questions and questions that remain insufficient at termination output an empty C_s and $y = 0$, respectively.

d) Complexity: The complexities of PPR, truncated path search, and greedy selection with cached statistics in each round are $O(T_p(|V| + |E|))$, $O(|Z_r|(|V| + |E|))$, and $O(|U_r|B_{\text{blk}})$, respectively, where $|Z_r| \leq (r + 1)K_{\text{seed}}$. The number of paths, the reranking pool, and the support-control pool all have fixed upper limits. Online cost is dominated by at most R_{max} rounds of sparse propagation, fixed-size reranking, and generation calls.

V. EXPERIMENTS

A. Experimental Settings

a) Datasets and Evaluation: The experiments use the full Qasper validation set (281 papers and 1005 questions) [44] and a fixed sample of 1000 questions drawn from the HotpotQA distractor validation set with random seed 42 [45]. Qasper covers full scientific papers, and each system submits paragraph-level evidence using a dynamic top- k strategy without access to gold answers or evidence; HotpotQA requires cross-document reasoning and sentence-level supporting facts.

The former reports Answer F1 and Evidence Precision, Recall, and F1, while the latter reports answer Exact Match (EM) and F1, Supporting Fact (SP) Precision, Recall, and F1, and Joint F1. All metrics are computed per question according to the official protocols and then macro-averaged. When multiple reference evidence sets exist, the three evidence metrics are taken from the same annotation set that maximizes Evidence F1; therefore, the F1 values in the table are not the harmonic means of the macro-averaged Precision and Recall.

b) Compared Methods: The compared methods include the weak-information lower bounds Abstract-only and Lead-only; the sparse, dense, fusion, and reranking methods BM25, Dense, Hybrid Reciprocal Rank Fusion (RRF), and BM25+Reranker [11–14]; the hierarchical or long-context methods RAPTOR and LongRAG [29,30]; the graph-enhanced methods HippoRAG 2 and KG²RAG [6, 7]; and the iterative methods IRCot and ReAct [18, 19]. All methods use the same document input and generation settings, and ReAct accesses only the fixed local corpus.

c) Implementation Settings: Atomic content is segmented by sentence or table row. The maximum lengths of local patches and section overviews are 1800 and 900 characters, respectively; each document outline retains at most 12 titles, and at most 8 entities or terms are extracted from each atomic block. A semantic edge is created when at least 2 terms are shared or the overlap ratio in (3) is at least 0.12. The weights of forward/reverse citation, parent-child, same-section, and section-membership edges are 0.95/0.70, 0.80, 0.35, and 0.90, respectively. All methods generate answers with Qwen/Qwen3-VL-8B-Instruct (temperature 0.1) and use Qwen/Qwen3-Embedding-0.6B and BAAI/bge-reranker-v2-m3. Seed retrieval retains the top 40/50 BM25/dense candidates (with $K_{\text{seed}} \leq 90$ after merging and $\beta = 0.35$); PPR uses $\alpha = 0.15$ and $T_p = 50$ and retains the top 60 candidates, while the top 50 atomic candidates are reranked with $\eta = 0.75$. The remaining main parameters are $K_{\text{path}} = L_{\text{path}} = 3$, $B_{\text{blk}} = 10$, $B_{\text{tok}} = 4000$, an upper limit of 2 auxiliary nodes, and $R_{\text{max}} = 2$. All settings are fixed before final evaluation and used for both datasets.

B. Main Experimental Results

Table I presents the main experimental results. On Qasper, CoSE-RAG achieves an Answer F1, Evidence Precision, Evidence Recall, and Evidence F1 of 45.44%, 47.34%, 57.26%, and 48.79%, respectively, exceeding the strongest baseline for each metric by 1.63, 2.28, 5.43, and 3.41 percentage points. On HotpotQA, its Answer EM, Answer F1, SP Recall, and SP F1 are 62.60%, 76.36%, 80.06%, and 67.52%, respectively, corresponding to improvements of 3.10, 1.63, 19.90, and 4.08 percentage points. All these metrics are the best, indicating that evidence-relation modeling and sufficiency feedback can jointly improve distributed-evidence coverage and answer quality.

a) Inference Cost: Table II shows that CoSE-RAG consumes 2331.0/1072.1 tokens per question and 8.08/6.37 seconds per question on Qasper and HotpotQA, respectively. Its

TABLE I
MAIN EXPERIMENTAL RESULTS ON THE TWO DATASETS (%). THE BEST AND SECOND-BEST RESULT IN EACH COLUMN ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY.

Category	Method	Qasper				HotpotQA					
		Ans. F1	Evi. P	Evi. R	Evi. F1	Ans. EM	Ans. F1	SP P	SP R	SP F1	Joint F1
Weak-information lower bound	Abstract-only	17.83	10.85	10.85	10.85	—	—	—	—	—	—
	Lead-only	—	—	—	—	34.40	41.12	21.00	20.32	19.01	10.58
Sparse, dense, and fusion retrieval	BM25 [11]	39.94	30.76	35.06	31.01	46.50	56.95	57.00	46.40	47.98	31.67
	Dense [12]	39.63	37.55	40.88	37.11	45.60	56.75	55.40	47.54	48.08	31.41
	Hybrid RRF [13]	40.68	37.08	41.60	36.96	47.80	59.13	60.15	50.13	51.37	34.26
	BM25+Reranker [14]	41.15	36.62	44.69	38.04	54.90	68.84	66.14	58.22	58.87	44.03
Hierarchical and long-context	RAPTOR [29]	40.38	38.33	42.00	38.03	48.50	61.42	51.50	46.69	45.64	31.16
	LongRAG [30]	<u>43.81</u>	<u>45.06</u>	<u>51.83</u>	<u>45.38</u>	51.40	65.36	69.73	<u>60.16</u>	62.03	44.69
Graph-enhanced retrieval	HippoRAG 2 [7]	39.97	27.68	40.42	30.33	<u>59.50</u>	74.52	83.88	55.08	<u>63.44</u>	50.23
	KG ² RAG [6]	37.38	38.28	42.83	37.74	59.00	<u>74.73</u>	<u>82.73</u>	50.06	59.53	46.77
Iterative retrieval and reasoning	IRCoT [18]	34.39	31.09	38.28	32.04	42.70	53.53	61.11	50.75	52.72	34.26
	ReAct [19]	25.80	14.71	24.08	16.34	51.20	64.49	41.50	56.11	45.56	32.08
Proposed method	CoSE-RAG	45.44	47.34	57.26	48.79	62.60	76.36	61.12	80.06	67.52	<u>49.88</u>

TABLE II
AVERAGE INFERENCE COST ON THE TWO DATASETS. THE BEST AND SECOND-BEST RESULT IN EACH COLUMN ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY.

Method	Qasper		HotpotQA	
	Tokens/Q. ↓	Sec./Q. ↓	Tokens/Q. ↓	Sec./Q. ↓
Abstract-only	403.8	3.00	—	—
Lead-only	—	—	807.92	5.07
BM25	1849.3	6.19	850.5	11.22
Dense	<u>1445.9</u>	9.58	<u>817.8</u>	10.48
Hybrid RRF	1703.8	7.49	835.4	10.22
BM25+Reranker	1896.6	9.56	847.5	5.52
RAPTOR	1542.8	6.85	1595.9	7.26
LongRAG	3760.3	5.30	1706.8	4.05
IRCoT	2175.4	5.25	1005.5	6.67
ReAct	9703.6	9.62	7360.7	7.87
HippoRAG 2	4483.2	15.61	4780.3	17.17
KG ² RAG	1623.6	<u>3.74</u>	1769.8	<u>4.78</u>
CoSE-RAG	2331.0	8.08	1072.1	6.37

token cost is lower than those of several hierarchical, graph-enhanced, or iterative baselines, but neither the time nor token metric is the best, placing its overall cost in the middle range. Although Abstract-only and Lead-only have the lowest costs, their task performance is substantially weaker, so they serve only as lower-bound references.

C. Ablation Study

Table III separately removes the multi-granularity seeds, the three edge types, the typed weights, the budgeted selector, and the sufficiency verifier. The final three variants use uniform edge weights, select top- k nodes by bridging score, and directly accept the first-round evidence, respectively.

All ablation variants show performance decreases. Removing the budgeted selector has the largest effect, reducing Qasper Answer F1/Evidence F1 by 6.81/4.22 percentage points and HotpotQA Answer F1/SP F1/Joint F1 by 4.77/4.36/5.09 percentage points. Qasper is more sensitive to

hierarchical and contextual edges, whereas removing semantic edges on HotpotQA reduces SP Recall and Joint F1 by 4.57 and 4.09 percentage points, reflecting the different dependence of the two tasks on section structure and cross-document semantic connections. Removing the sufficiency verifier also reduces Evidence F1/Joint F1 on the two datasets by 1.43/1.54 percentage points, confirming the effect of targeted supplementary retrieval.

D. Retrieval-Trajectory Case Analysis

Table IV presents two real retrieval trajectories. In the Qasper case, the first round does not find evidence that directly lists the datasets. Semantic bridging retrieves the IITB and ILCI English–Hindi parallel corpora in the second round, increasing Evidence F1 from 16.67% to 33.33% and yielding a final Answer F1 of 94.12%; BM25+Reranker instead mistakes language names for datasets. In the HotpotQA case, contextual and shared-term relations supplement the connection between Ross Pople and the London Festival Orchestra in the second round, increasing Supporting Fact F1 from 46.15% to 57.14% and identifying the correct person, whereas BM25+Reranker incorrectly outputs “Yes.” The two cases show that controlled expansion triggered by sufficiency diagnosis can supplement missing evidence while keeping the final supporting evidence traceable.

E. Parameter Sensitivity Analysis

We use a one-variable-at-a-time analysis for $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$, $K_{\text{path}} \in \{0, 1, 3, 5, 7\}$, $B_{\text{blk}} \in \{4, 6, 8, 10, 12\}$, and $R_{\text{max}} \in \{1, 2, 3, 4\}$, while keeping the remaining parameters at their main-experiment settings.

As shown in Fig. 2, Qasper Evidence F1 and HotpotQA Joint F1 are both highest at $\alpha = 0.15$, $K_{\text{path}} = 3$, $B_{\text{blk}} = 10$, and $R_{\text{max}} = 2$. Parameters that are too small limit evidence connection, whereas further increases introduce redundancy,

TABLE III
ABLATION RESULTS OF COSE-RAG (%). THE BEST AND SECOND-BEST RESULT IN EACH COLUMN ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY.

Model Variant	Qasper				HotpotQA					
	Ans. F1	Evi. P	Evi. R	Evi. F1	Ans. EM	Ans. F1	SP P	SP R	SP F1	Joint F1
w/o multi-granularity seeds	40.17	45.58	54.13	46.12	61.30	74.91	59.36	76.87	64.98	47.63
w/o context edges	39.86	44.47	53.09	45.07	61.10	74.72	58.85	77.21	64.96	47.49
w/o semantic edges	39.92	45.19	54.71	46.38	60.10	73.41	57.86	75.49	63.67	45.79
w/o hierarchy edges	39.08	43.27	53.58	44.71	61.60	75.11	59.57	<u>78.31</u>	65.89	<u>48.42</u>
w/o typed weights	40.31	44.68	54.29	45.91	61.40	74.89	58.96	77.48	65.17	47.81
w/o budgeted selector	38.63	42.61	55.19	44.57	58.20	71.59	55.74	77.63	63.16	44.79
w/o sufficiency verifier	<u>40.61</u>	<u>46.08</u>	<u>55.31</u>	<u>47.36</u>	<u>61.70</u>	<u>75.29</u>	<u>59.87</u>	78.11	<u>66.09</u>	48.34
Full CoSE-RAG	45.44	47.34	57.26	48.79	62.60	76.36	61.12	80.06	67.52	49.88

TABLE IV
REAL RETRIEVAL TRAJECTORIES ON QASPER AND HOTPOTQA. ONLY REPRESENTATIVE RETRIEVED BLOCKS ARE LISTED FOR EACH ROUND; ALL EVIDENCE IDENTIFIERS CAN BE TRACED TO THE SOURCE TEXT THROUGH PROVENANCE MAPPING; AUXILIARY NODES PARTICIPATE ONLY IN EXPANSION AND ARE NOT USED AS FINAL SUPPORTING EVIDENCE.

Stage	Qasper Case	HotpotQA Case
Question / Gold	Question: Which dataset(s) do they experiment with? Reference answer (normalized): IITB English-Hindi parallel corpus; ILCI English-Hindi parallel corpus	Question: Ross Pople worked with this French dominant figure of the post-war classical music world. Gold answer: Pierre Louis Joseph Boulez
BM25+Reranker	[4] (Introduction) Deep Learning approaches have achieved impressive results on various NLP tasks BIBREF0 , BIBREF1 , BIBREF2 and have become the de facto approach for... [2] (Abstract) Transfer learning approaches for Neural Machine Translation (NMT) train a NMT model on the assisting-target language pair (parent model) which is lat... Answer: Bengali, Gujarati, Marathi, Malayalam, and Tamil; Evidence-set F1=0.00%.	[36] (Pierre Boulez) He was one of the dominant figures of the post-war classical music world. [3] (London Festival Orchestra) In 1980 it was incorporated as an independent performing orchestra under Ross Pople. Answer: Yes; Evidence-set F1=33.33%.
Round 1	[21] (Experimental Setup) In this section, we describe the languages experimented with, datasets used, the network hyper-parameters used in our experiments. [6] (Introduction) To see how word order divergence can be detrimental, let us consider the case of the standard RNN (Bi-LSTM) encoder-attention-decoder architecture BI... First-round Evidence-set F1=16.67%.	[35] (Pierre Boulez) Pierre Louis Joseph Boulez CBE () ; 26 March 1925 – 5 January 2016) was a French composer, conductor, writer and organiser of institutions. [36] (Pierre Boulez) He was one of the dominant figures of the post-war classical music world. First-round Evidence-set F1=46.15%.
Diagnosis	$\mathcal{M}$: relevant_evidence, too_noisy; $\mathcal{L}$: semantic; ξ_{next} : expand_semantic_bridge.	$\mathcal{M}$: table_or_caption_context; $\mathcal{L}$: context; ξ_{next} : expand_table_caption.
Round 2	[25] (Datasets) For training English-Hindi NMT systems, we use the IITB English-Hindi parallel corpus BIBREF22 (INLINEFORM0 sentences from the training set) and the ILCI English-Hindi parallel corpus (INLINEFORM1... [27] (Network) We use OpenNMT-Torch BIBREF24 to train the NMT system. We use the standard sequence-to-sequence architecture with attention BIBREF13 . We use an encoder which contains two layers of bidirectional LST... [23] (Languages) We experimented with English INLINEFORM0 Hindi translation as the parent task. English is the assisting source language. Bengali, Gujarati, Marathi, Malayalam and Tamil are the primary source languag... Relevant bridges: semantic/shared_terms, context/child_parent, hierarchy/belongs_to_section, semantic/mentions_entity.	[26] (Motonari Iguchi) Motonari Iguchi (Iguchi Motonari) (17 May 1908 in Tokyo – 29 September 1983 in Tokyo) was a Japanese pianist and educator. [39] (Ross Pople) He is the principal conductor of the London Festival Orchestra. Relevant bridges: semantic/shared_terms, context/same_section, context/next, context/previous.
Final Output	Answer: IITB English-Hindi parallel corpus and ILCI English-Hindi parallel corpus; Supporting block: [25]; Answer F1=94.12%; Evidence-set F1: 16.67%→33.33%.	Answer: Pierre Boulez; Supporting blocks: [36, 40, 38]; Evidence-set F1: 46.15%→57.14%.

indicating that moderate propagation, path connection, and retrieval rounds balance evidence coverage and noise.

F. Discussion of Results

CoSE-RAG achieves the best results on all four Qasper metrics and on HotpotQA Answer EM, Answer F1, SP Recall, and SP F1, but not on HotpotQA SP Precision or Joint F1.

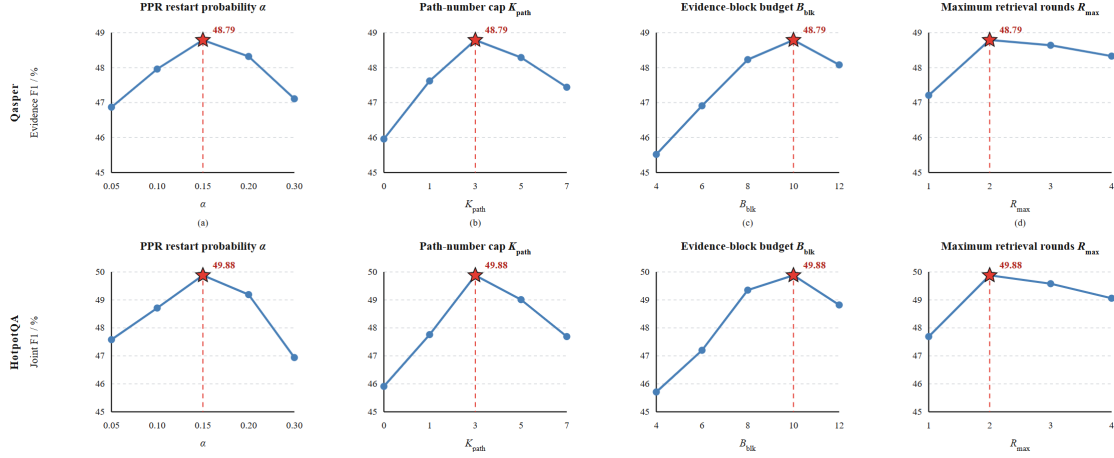


Fig. 2. Sensitivity analysis of different hyperparameters on Qasper and HotpotQA. The top row reports Evidence F1 on Qasper, and the bottom row reports Joint F1 on HotpotQA; red stars and vertical dashed lines indicate the parameter values used in the main experiments.

This reflects a trade-off between coverage and precision: supplementary retrieval improves supporting-fact recall but may retain a small amount of unnecessary evidence, while Joint F1 requires both the answer and supporting facts to be correct for the same question. Overall, the method’s gains mainly come from sufficient retrieval and integration of distributed evidence.

VI. CONCLUSION

This paper presents CoSE-RAG, which organizes heterogeneous document content in a provenance-preserving multi-view graph and constructs compact supporting contexts through requirement-aware seed retrieval, typed evidence bridging, dual-budget selection, and sufficiency-driven re-retrieval. Experiments on Qasper and HotpotQA show the best results on all four Qasper metrics and on HotpotQA Answer EM, Answer F1, Supporting Fact Recall, and Supporting Fact F1; the ablation results confirm the contributions of the principal components, with budgeted selection having the largest effect. The current method operates on traceable text obtained after document parsing; extending it to visual evidence and larger-scale evaluation remains future work.

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